

# Mohammed Sameer Pasha

+91-93929-74781 | [mohammedsameerpasha128@gmail.com](mailto:mohammedsameerpasha128@gmail.com) | Hyderabad, Telangana, India  
[linkedin.com/in/sameer-pashaaa](https://www.linkedin.com/in/sameer-pashaaa) | [github.com/sameerpashaaa](https://github.com/sameerpashaaa) | [sameerpashaaa.github.io/portfolio](https://sameerpashaaa.github.io/portfolio)

## Professional Summary

Generative AI Engineer with 1+ year of production experience building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, and multi-agent AI architectures. Expertise in Python, PyTorch, LangChain, and full-stack development (Django, FastAPI), with proven ability to deploy scalable machine learning models and AI-driven enterprise solutions. Skilled in natural language processing, computer vision, vector databases, and cloud-based model deployment. Delivered AI systems serving 50+ concurrent users and processing 5,000+ documents with sub-3-second response times.

## Technical Skills

**Programming Languages:** Python, SQL, JavaScript, Bash, Git  
**Generative AI & LLMs:** LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, CrewAI, OpenAI API, Groq, Mistral, Chain-of-Thought Reasoning, Function Calling, LLM Orchestration  
**Machine Learning:** Supervised Learning, Model Training & Evaluation, Feature Engineering, Scikit-Learn, Hyperparameter Tuning, Cross-Validation, Model Optimization, Data Preprocessing  
**Deep Learning:** PyTorch, TensorFlow, Convolutional Neural Networks (CNNs), Transfer Learning, Model Fine-tuning, HuggingFace Transformers, YOLO, Computer Vision  
**Natural Language Processing:** NLP, Sentence Transformers, Embedding Models, Intent Classification, Named Entity Recognition, Text Classification  
**Speech AI:** Speech-to-Text, Text-to-Speech, LiveKit (Voice Agents), Real-Time Audio Processing, Bilingual NLU  
**Vector Databases:** FAISS, ChromaDB, Semantic Search, Embedding Pipelines, Similarity Search  
**Web Frameworks:** Django, Flask, FastAPI, Streamlit, REST APIs, WebSockets, Microservices  
**Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Redis (Caching), Database Optimization, Query Performance Tuning  
**MLOps & Deployment:** Docker, Linux, Git, CI/CD, Model Serving, API Development, Model Versioning, Oracle Cloud Infrastructure, Google Colab, Jupyter Notebook  
**Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Data Visualization, ETL Pipelines, Statistical Analysis, Exploratory Data Analysis (EDA)  
**Computer Vision:** OpenCV, Image Processing, Object Detection, OCR (Optical Character Recognition), YOLOv8, EasyOCR, Preprocessing Techniques  
**Development Tools:** VS Code, Postman, GitHub, Kaggle, Jupyter, Git Version Control  
**Soft Skills:** System Architecture Design, Client Communication, Requirement Gathering, Technical Documentation, Agile Development, Cross-Functional Collaboration, Problem Solving

## Professional Experience

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|-----------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Metaphasic Life Sciences</b><br><i>Software Engineer – AI &amp; Full Stack Development</i> | Hyderabad, India<br>March 2024 – Present |
|-----------------------------------------------------------------------------------------------|------------------------------------------|

- Architected and deployed a multi-tenant Enterprise Resource Planning (ERP) system using Django, PostgreSQL, and REST APIs, serving 5 departments with 50+ concurrent users, reducing operational processing time by 40% through automated workflow orchestration and role-based access control.
- Engineered HIPAA-aligned Electronic Health Records (EHR) system using Django and encrypted PostgreSQL, enabling 3 healthcare clinics to digitize 10,000+ patient records with audit logging, appointment scheduling, and prescription management, improving record retrieval time by 90%.
- Delivered real-time inventory management system for 200+ room hotel client using Flask, MySQL, and RESTful APIs, tracking 5,000+ SKUs across 8 departments, reducing stock-outs by 65% and cutting manual reconciliation time from 8 hours to 20 minutes through automated threshold alerts.
- Developed and launched responsive corporate website achieving 95+ PageSpeed score and 300% increase in inbound lead generation within 3 months through SEO optimization and conversion-focused UI/UX design.
- Implemented role-based authentication and authorization system using JWT tokens and Django permissions framework, securing 6 distinct user role hierarchies with granular module-level access control.
- Optimized database query performance by implementing indexed lookups and Redis caching, reducing average API response time from 1.8s to 240ms for high-traffic endpoints serving 1,000+ daily requests.

## Projects

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|------------------------------------------------------------------------------------|----------|
| <b>StackMind</b>   <i>Production RAG System with Multi-Agent LLM Orchestration</i> | Dec 2024 |
|------------------------------------------------------------------------------------|----------|

- Technologies:* LangChain, RAG, Groq, FAISS, CrewAI, Python, Streamlit
- Engineered production-grade Retrieval-Augmented Generation (RAG) chatbot using LangChain, Groq LLMs, and FAISS vector database, processing 5,000+ technical documents with 89% answer relevance score and sub-3-second response latency for multimodal queries (text, code, diagrams, voice).

- Implemented chain-of-thought prompting with ReAct framework for complex query decomposition, improving answer accuracy by 34% on multi-hop reasoning tasks and enabling transparent step-by-step explanations for technical problem-solving workflows.
- Architected multi-agent AI system using CrewAI framework with specialized agents (Researcher, Planner, Code Analyst, Synthesizer) orchestrating parallel task execution, achieving 40% faster complex query resolution through role-based delegation, shared memory context, and real-time API integration.
- Optimized vector embedding pipeline using Sentence Transformers and semantic chunking strategies, reducing context retrieval time by 60% through sliding-window segmentation with 200-token overlap for long-form technical documentation.
- Built Streamlit web interface with real-time streaming responses, session memory persistence, and multimodal input handling (file upload, voice recording, code snippets), supporting 50+ concurrent user sessions with stateful conversation tracking.

**Restaurant AI** | *Real-Time Voice Agent with Bilingual NLU & LLM Integration*

Nov 2024

- *Technologies: LiveKit, Mistral API, SQLite, Python, Speech Processing*
- Developed enterprise-ready voice AI agent using LiveKit for real-time speech processing (150ms latency), Mistral LLM for intent classification and dialogue management, supporting English/Hindi bilingual conversations with 92% intent recognition accuracy across 20+ reservation scenarios.
- Engineered intelligent reservation system with Mistral API integration for context-aware dialogue flow, managing 15-table availability matrix, party-size optimization, and conflict resolution logic, reducing average booking time from 3 minutes (phone) to 45 seconds through natural language automation.
- Implemented full-duplex voice pipeline with LiveKit's WebRTC streaming (speech-to-text + text-to-speech), achieving 150ms round-trip latency and 94% transcription accuracy (English/Hindi), with intelligent seating recommendation engine integrating OpenWeatherMap API for indoor/outdoor preference optimization.
- Designed SQLite database schema with normalized tables for reservations, customer history, and table status, implementing transaction-safe CRUD operations and concurrent booking conflict detection with row-level locking.

**ANPR System** | *End-to-End License Plate Detection with YOLOv8 & OCR Pipeline*

May 2024

- *Technologies: YOLOv8, OpenCV, EasyOCR, PyTorch, Python, Computer Vision*
- Engineered end-to-end Automatic Number Plate Recognition (ANPR) pipeline using YOLOv8 for real-time detection and EasyOCR for text extraction, trained on 15,000+ annotated images (Kaggle dataset) achieving 91% plate localization mAP@0.5 and 87% character recognition accuracy at 18 FPS on CPU.
- Developed adaptive preprocessing pipeline with Gaussian blur, Otsu's binarization, morphological transformations, and CLAHE enhancement for low-light/skewed plates, boosting OCR accuracy from 62% to 87% (+25 percentage points) and reducing false positives by 40% through contour area filtering.
- Optimized inference pipeline with frame skipping (process every 3rd frame), batched preprocessing, and ROI-focused detection zones, reducing per-frame processing time from 5.2s to 1.8s (65% improvement) and achieving 18 FPS throughput for real-time video stream analysis.
- Implemented real-time visualization dashboard using OpenCV and Streamlit, displaying bounding boxes, OCR confidence scores, and recognized plate numbers with 50-frame history tracking and CSV export functionality for analytics and validation.

## Education

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|---------------------------------------------------------------|-----------------------------|
| <b>Muffakham Jah College of Engineering and Technology</b>    | Hyderabad, India            |
| <i>B.Tech in Artificial Intelligence and Machine Learning</i> | <i>Aug 2022 – Aug 2026</i>  |
| <b>Sri Chaitanya Junior College</b>                           | Hyderabad, India            |
| <i>Intermediate (MPC) – Mathematics, Physics, Chemistry</i>   | <i>June 2020 – May 2022</i> |

## Certifications

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|-------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Oracle Cloud Infrastructure 2024 AI Foundations Associate</b>                                                        | Sept 2024 |
| <i>Certified in OCI AI fundamentals, cloud AI services, deployment workflows, and responsible AI frameworks</i>         |           |
| <b>AWS Solutions Architecture Virtual Experience</b>                                                                    | July 2024 |
| <i>Designed scalable AWS Elastic Beanstalk architecture with auto-scaling, load balancing, and cloud best practices</i> |           |
| <b>Deloitte Technology Job Simulation</b>                                                                               | June 2024 |
| <i>Built performance monitoring dashboard and delivered technical recommendations to stakeholders</i>                   |           |
| <b>100 Days of Code: Python Pro Bootcamp (Udemy)</b>                                                                    | May 2024  |
| <i>Comprehensive Python development course covering data structures, OOP, web frameworks, and automation</i>            |           |